Sizing and Configuration Decisions

AMERICAN UNIVERSITY OF THE MIDDLE EAST

DOC AUM-DBF-27-DB-001
REV A / 14 OCT 2026
PAGE 01 / 06

1.0 PURPOSE

The configuration locks on 17 Oct and the numbers freeze on 20 Oct. This brief puts the evidence for every open sizing and configuration decision in one place. For each decision it gives the options, the numbers and the risks, a recommendation, and what would change it.

Status. Nothing here is decided. Each recommendation is the analysis's, written by Claude; the team decides in Section 8.0.

Basis. Scripts 1–6 and the Table 06 trade. The model includes the deployment-and-recovery (D&R) mechanism (§3.1.2: installed for all missions). It is calibrated on 452 values verified against the 2021–26 top-3 AIAA reports and 513 fly-off results rows.

BRIEF AT A GLANCE

DECISIONS	DUE
12 open	**17 Oct**
WING	PAYLOAD UNIT
0.9 m²	**4** kg
MEAN TOTAL	P(M2 FEASIBLE)
6.33 of 7.5	**98** %

2.0 HOW TO READ THIS BRIEF

Score. The figure of merit is the total mission score $M_1 + M_2 + M_3 + GM$ (at most 7.5), against the *expected* field. That field is 75 % of per-mission specialist maxima anchored on the 2021–26 AIAA reports.

Uncertainty. Every design is scored over the same 120 random draws of the uncertain inputs (common random numbers):

- resampled from measured data: lap-time realism (24 flight pairs, median ×1.15), pack specific energy (9 packs) and airframe mass (fit scatter, ±31 % at 1 sd);
- over estimate ranges: drag, efficiency, CLmax, air density and container mass.

"Mean" is the average over the draws. P10 and P90 bound the worst and best tenth.

Evidence grades.

- *Verified*: quoted from a primary source and machine-checked.
- *Reseller*: a retail listing only.
- *Estimate*: a reasoned range, sampled in the Monte Carlo.

What the model does not price: take-off and landing risk, structural failure, build schedule and pilot skill. These enter through the risk register (Section 7.0) and the team's judgment.

3.0 DECISION SUMMARY

Table 01 lists every decision with the recommended option, the confidence in it, and what would change it.

The airfoil is not reopened here: the proposal's Fig. 05 XFoil study over 13 sections picks the SD7037 for the race envelope, and the team's airfoil matrix is dropped (its arithmetic flips its own winner).

4.0 SIZING (D1–D5)

4.1 Wing Area and Planform (D1)

The robust map (Fig. 01) scores every wing area against every payload unit over the 120 draws.

Best point: 0.9 m² with a 4 kg unit. Mean 6.33; P(M2 feasible) 98 %. The plateau spans 0.8–1.0 m², and the candidates rank as follows (same draws, each at its best unit):

- 1.0 m²: 6.29;
- 0.8 m²: 6.23 (1.0 m² is better in 65 % of draws);

TABLE 01 Open decisions and the decision record. “Rec.” is the analysis’s recommendation, not a decision; confidence reflects the evidence, not the importance. The team fills the last column (Section 8.0).

ID	DECISION	REC.	CONFIDENCE	CHANGES IF	TEAM DECISION
D1	Wing area, planform	Straight, flapped, 0.9 m² (0.9–1.0)	High	Stall below 13 m/s (then 1.0–1.3 m²)	
D2	Accepted 1 g stall at M2 weight	14 m/s (167 N/m²)	Pilot’s call	Pilot’s number; +0.13–0.15 per m/s above 14	
D3	Payload unit and sensor	4 kg unit (3.4 kg sensor), simulators as mass allows; sensor declared at tech	High	First weigh-in far below the fit	
D4	Mission 1	Waive (credited by M2)	High	A Q&A ruling on the credit	
D5	Drop height	Design 1.524 m; declare after TC-02	Medium	TC-02 fails at 1.524 m	
D6	Wing position	High wing	Medium	CAD favours a low-wing exit or loading path	
D7	Tail	Conventional	Medium	CAD clearance check fails	
D8	Landing gear	Tricycle, fixed, faired	Medium	Pilot prefers a taildragger	
D9	Propulsion layout	Single tractor	Low (pre-study)	The propulsion study (next)	
D10	Sensor-bay closure	Door, unless an open slot costs under 2–4 cm²	Low	WP-01 CFD on the slot	
D11	Body shape	From packaging: two units, reel, sensor bay	Deferred	TC-02 sets the container	
D12	D&R architecture	A, closed-loop drum	Medium-high	TC-05 bench counts	

- 0.6 m²: 5.78;
- the 0.414 m² placeholder: 5.09;
- the delta: 4.60.

The straight wing beats both the placeholder and the delta in 99 % of draws.

Why not 0.8 m². With the reel’s 0.32 kg on board, a 10 % smaller wing loses the payload step (tornado: −0.24). Pick 0.9 m² or slightly above, not the bottom of the plateau.

Planform. Straight and flapped; the delta is only a foil. The taper ratio comes from XFLR5 (span efficiency, stall progression) and manufacturing; the team’s planform matrix is dropped because four of its five criteria were copied from the wing-position matrix.

4.2 Accepted Stall Speed (D2)

The stall bound fixes the M2 wing loading, $W/S = \rho V_s^2 C_{L,\max}/2$: 167 N/m² at 14 m/s ($\rho$ 1.064, $C_{L,\max}$ 1.6). It is the model’s biggest single lever (Fig. 02, Table 02):

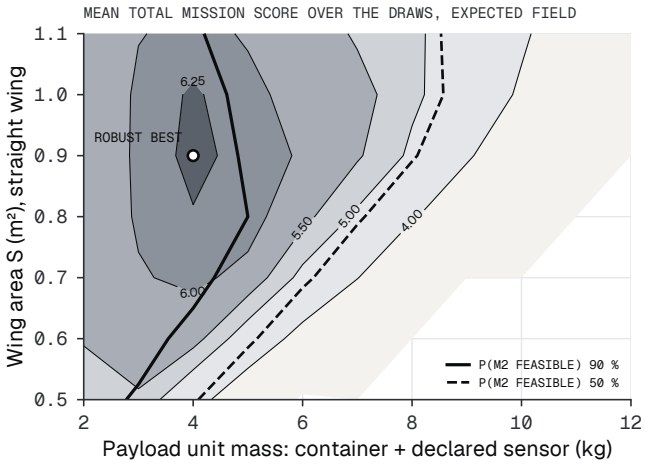


FIG. 01 Robust design map: mean total score over 120 draws, straight wing, 14 m/s stall, reel included. Solid and dashed lines: P(M2 feasible) of 90 % and 50 %. Source: `s6_report.txt`.

every 1 m/s above 14 adds 0.13–0.15 to the mean, and 11 m/s costs 1.4.

The gain is not free. The model has no take-off distance limit and does not price landing risk.

- In 2025 the two heavy top-3 designs (UNSW,

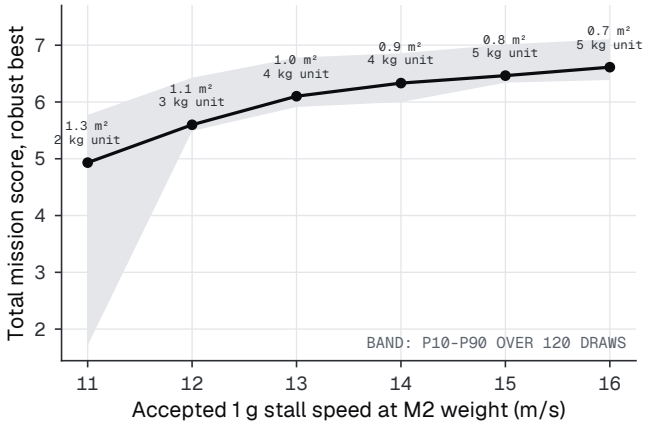


FIG. 02 Robust best design for each accepted stall speed. Wing area grid 0.6–1.4 m², units 2–9 kg. Band: P10–P90 over the 120 draws. Source: `db01_report.txt`.

TABLE 02 Stall-bound sweep (`db01_report.txt`). W/S at $C_{L,\max}$ 1.6, ρ 1.064.

STALL M/S	W/S N/M²	WING M²	UNIT KG	MEAN	P(M2)
11	103	1.3	2	4.93	86 %
12	123	1.1	3	5.60	91 %
13	144	1.0	4	6.10	93 %
14	167	0.9	4	6.33	98 %
15	192	0.8	5	6.46	95 %
16	218	0.7	5	6.61	98 %

ERAU) both scored zero in M2 at the fly-off, at 248–257 N/m² (about 17 m/s).

- UCLA 2025 completed every mission near 167 N/m².

Rec.: 14 m/s, unless the pilot signs off on 15 m/s (+0.13) after seeing the landing speed.

4.3 Payload Unit and Sensor Mass (D3)

Simulators must match the container plus the declared sensor within ±28.35 g (§3.1.3), so M2 payload comes in whole units. The declared sensor also sets GM. Table 03 shows the trade at 0.9 m².

The 4 kg unit has the best mean and the best P10.

- Heavier units raise the median and GM: 8 kg reaches the GM ceiling of 1.50.
- But M2 becomes infeasible whenever the airframe comes in heavy: P(M2) falls from 98 % to 52 % at 8 kg, and P10 collapses to the M1-plus-GM floor. An M2 failure also blocks M3.

Keep the option open. The declared sensor mass is fixed only at tech in April (§4.1). Size the bay for two container-sized units now, and choose the sensor mass after the first weigh-in and flight tests.

TABLE 03 Payload unit at 0.9 m², 14 m/s (`db01_report.txt`). Sensor = unit − 0.6 kg container. Mission parts are means.

UNIT KG	MEAN	P10	P(M2)	M2	M3	GM
2	5.62	5.25	99 %	1.48	2.43	0.72
3	6.07	5.61	99 %	1.47	2.72	0.88
4	**6.33**	**6.00**	**98 %**	**1.43**	**2.85**	**1.05**
5	6.15	2.25	88 %	1.29	2.64	1.22
6	5.96	2.37	80 %	1.18	2.40	1.38
8	4.89	2.50	52 %	0.84	1.55	1.50

4.4 Mission 1 Waiver (D4)

A successful M2 also credits M1, and attempts are scarce (2.4–2.9 official attempts per team that passed tech, 2022–26). At the field's 65 % success per attempt with three attempts, P(M3 completed) rises from 40 % flying M1 to 72 % waiving it (with four attempts: 70 % to 87 %). Every Script 6 number above assumes the waiver.

The cost: a failed M2 can no longer fall back on M1.

4.5 Drop Height (D5)

The GM score rises linearly with the declared height up to 1.524 m (60 in). Design the container for 1.524 m. Declare the largest height TC-02 passes cleanly, with the real sensor, before the design report.

5.0 CONFIGURATION (D6–D11)

Field practice does not pick a configuration. Of the 19 verified top-3 aircraft since 2021:

- 13 flew a high wing, 13 a tricycle gear, 16 a conventional tail and 12 a single motor (Fig. 03);
- no feature predicts fly-off success. Low wings completed M3 5 of 6 times against 4 of 9 for high wings ($p = 0.29$). Single motors completed 7 of 10 against 1 of 4 for twins ($p = 0.24$). Gear type: $p = 1.00$ (Fisher exact).

So each row below is decided by physics, packaging and risk. The teammates' matrices were audited (arithmetic, evidence) and re-weighted with the Script 6

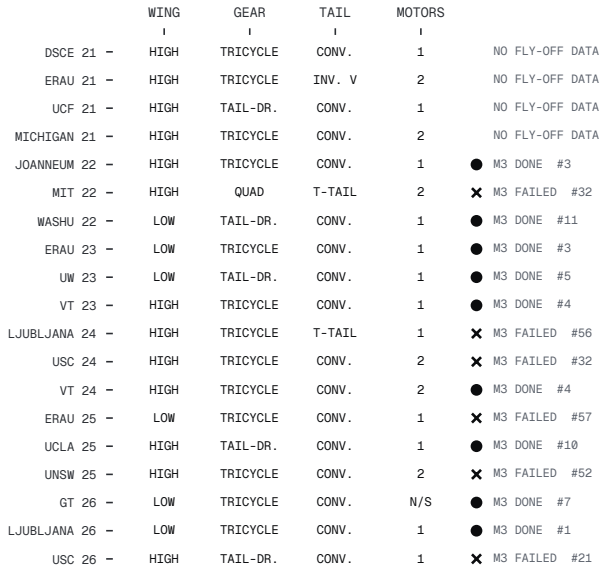


FIG. 03 Configurations of the 19 verified top-3 aircraft, 2021–26, with fly-off outcomes. Source: `data/dbf_config.csv`.

TABLE 04 Team matrices after the audit and Script 6 weights: P(option ranks first) under weight and score perturbation (`trades_report.txt`).

MATRIX	RESULT
Wing position	High 95 %, low 5 %, mid 0 %
Tail	Conventional 78 %, T-tail 14 %, twin-boom 8 %
Landing gear	Tricycle 76 %, taildragger 24 %
Fuselage structure	Box-beam 100 % (manufacturing plan)
Airfoil, planform	Dropped (errors, copied criteria)

sensitivities. Table 04 gives each option's P(first) over 5,000 perturbations.

D6 Wing position: high wing (medium). It wins 95 % of the audited trade. That trade had to correct the teammates' low-wing payload score: four low-wing top-3 designs (WashU 2022, UW 2023, ERAU 2023 and 2025) chose low wing *for* payload access. Decide finally on three things: the sensor exit path, the staging access through a top or bottom hatch (Q&A pending on §3.3.2 hatches), and the pilot.

D7 Tail: conventional (medium).

- The as-made matrix chose a T-tail on "line clearance". But the tow line hangs 31–86° below horizontal (0.5–5 kg sensor), so clearance is set by the exit location, not the tail type.
- A T-tail's fin carries the stabiliser loads, which makes it heavier and more flutter-prone.
- Check the clearance in CAD once the exit is placed.

D8 Gear: tricycle, fixed and faired (medium).

- A retract system must weigh under 120–240 g to pay for itself. That threshold is the fixed gear's drag area (9.9 cm² measured on GT 2026, 19.4 cm² predicted for Joanneum 2022) × 12.3 g per cm².
- It would save at most 2.6–5.1 s of M2 (about 0.01 points), against the risk of a gear-up landing.
- Fairings cut gear drag (UCLA 2025: 7 %).

D9 Propulsion layout: single tractor (low, provisional).

- A nose tractor keeps the propeller disc away from the tow line, whose slack moments (deploy, stow) are the SR-10 risk.
- The field favours single motors (12 of 19), and twins completed M3 only 1 of 4 times.
- The propulsion study (next: APC and UIUC data, motor model, 12S against 8S) confirms or overturns this.

D10 Sensor-bay closure (low). §3.1.1: an opening left uncovered in M3 must stay uncovered in M1 and M2.

- At the exchange rate of 10 cm² of drag area per 123 g, a door mechanism of 30–50 g (estimate) pays for itself if the open slot's drag area exceeds 2.4–4.1 cm².
- A slot sized for a can-sized dart with fins likely exceeds that, but no verified cavity-drag figure exists. WP-01 CFD should compute it.
- A door adds one failure mode (it fails to open, so no deploy), which the latch test covers.

D11 Body shape (deferred). Through drag, the body moves the score only by hundredths (10 cm² of drag area = 123 g). Its mass and its payload bay decide it: two container-sized units, the reel at the M3 location and the sensor bay. TC-02 fixes the container size. A CAD packaging study then gives the small-

est body, and any outer form (whale-like, MQ-4C-like, tube) is scored on the same drag-and-mass scale (`config_exchange.pdf`).

6.0 D&R ARCHITECTURE (D12)

Eight concepts were screened out against the rules or physics:

- a receiver on the drum (§3.2.1 external switch);
- a pulley block (77–153 s to deploy against a 20 s window);
- a boom, a wireless link and a two-line bridle (§3.1.1);
- a spring reel;
- a clock-spring lead (23 turns);
- a sail-winch servo (2.3× its torque).

Four candidates were sized with datasheet drives and compared pairwise on five criteria. Weights are the expected points each criterion can swing:

- 100 g of mechanism costs 0.046 points;
- 1 s of stow costs 0.002;
- 1 % more chance of losing the sensor costs 0.028.

So reliability decides it.

Rec.: A, closed-loop drum. Encoder, stop bead at the fairlead (sets the 2.74 m deploy length and carries lap loads into the airframe), slip ring and docking latch. It is the Condorcet winner and wins 95–98 % of the draws (Fig. 04). The open-loop drum (B) is the runner-up: it is easier to build and wins 29 % on the weighted table, but it ties A only if its own failure rates are below 46 % of the estimate.

Weak spot. The failure rates are engineering estimates. TC-05, a tow, reel and slip-ring bench card, must replace them with counted cycles.

Drive. The model carries the verified Pololu 37D 70:1 (0.32 kg mechanism, 12 s stow at 3.4 kg). A brushless servo (reseller data) would save about 0.12 kg.

7.0 RISK REGISTER

Table 05 lists the risks that can cost a mission, each with the test or analysis that retires it. Likelihood and consequence: L (low), M (medium), H (high).

8.0 DECISIONS REQUESTED

Table 01 is also the decision record: the team writes its choice in its last column.

- **By 17 Oct:** D1–D4, D6–D8 and D12. D1 waits on D2, so start with the pilot.
- **By 20 Oct:** D5, D9, D10 and D11. They wait on TC-02, the propulsion study and WP-01 CFD.

Only after sign-off do the targets enter `analysis/inputs/aircraft.toml` and the proposal; Table 06 of the proposal is wired after D12.

9.0 SOURCES

Analysis reports in `analysis/out/`:

- `s4_report.txt`: sizing;
- `s6_report.txt`: Monte Carlo, robust map, tornado;
- `db01_report.txt`: stall sweep, unit table, waiver;
- `config_trades_report.txt`: exchange rate, retract break-even;
- `trades_report.txt`: matrix audit;
- `dr_trade_report.txt`: Table 06.

Field and configuration data: `analysis/data/`, each value with its verbatim quote and page (`verify.py`: 0 failed).

Primary documents:

- AIAA DBF 2026–27 rules, draft of 2 Sep 2026 (latest official);
- AIAA competition summaries and top-3 design reports, 2021–26;
- manufacturer data for the D&R drives (Pololu, goBILDA, REV), read 13–14 Oct 2026.

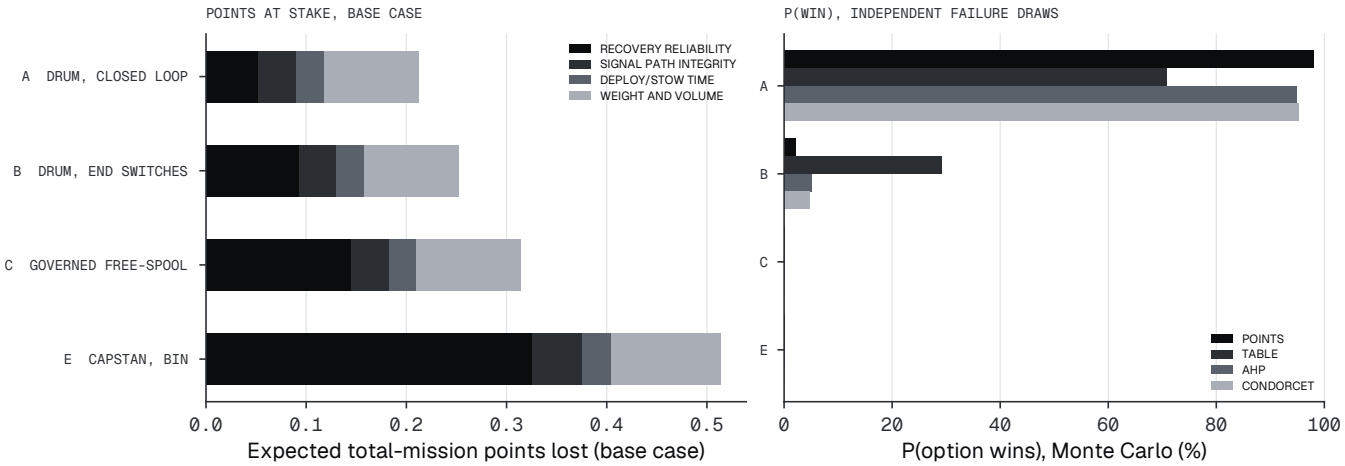


FIG. 04 D&R trade. Left: expected total-mission points lost by criterion, base case. Right: P(option wins) over 5,000 draws with each option's failure rates drawn independently, under four decision rules. A: closed-loop drum; B: open-loop drum; C: governed free-spool; E: capstan and bin. Source: `dr_trade_report.txt`.

TABLE 05 Risk register. The Spearman values are each input's correlation with the total score over the draws (`s6_report.txt`).

#	RISK	L	C	MITIGATION	RETIRED BY
R1	Airframe heavier than the fit (±31 % at 1 sd; Spearman −0.64) makes M2 infeasible	M	H	4 kg unit with optional simulators; weigh every part	CAD mass properties (WP-01 K5); first weigh-in
R2	Laps slower than modelled (median ×1.15, up to ×1.48; Spearman −0.68)	H	M	Calibrated model; margin on the M2 clock	GPS laps in flight test
R3	Stall too fast for reliable landings at full payload	M	H	14 m/s bound, flaps	Pilot sign-off; heavy-landing flight test
R4	Sensor or container fails the six GM drops (only one sensor)	M	H	Container for 1.524 m; GM after M3 (Q&A)	TC-02 with the real sensor
R5	Sensor not stowed for landing (jam, stall)	L–M	H	Closed-loop drum, stop bead, stall detection	TC-05 cycles at 3.4 kg
R6	Sensor unstable in tow (spins: M3 fails)	M	H	Fixed-geometry dart	TC-01 blower rig; CFD
R7	Lights not distinguishable in sun	M	H	Bright arrays, slow flash	TC-03
R8	Five-minute staging overrun (M3 sensor attach, M2 loading)	M	M	Tool-less attach, drilled crew	TC-04
R9	Tow line reaches the propeller or structure (SR-10)	L	H	Tractor layout; exit placed for clearance	CAD check; flight test
R10	Propulsion short of power or overheating at 100 A	M	H	Propulsion study with measured data	Thrust stand
R11	Rules read differently (GM timing, stow in window, hatches)	M	M	Conservative assumptions	Q&A round 1 (1 Nov)
R12	Parts late (PR-01 motor and prop numbers were due 14 Oct)	H	M	Order right after the propulsion study	PR-01 filed